Gain selection with a third-order actuator

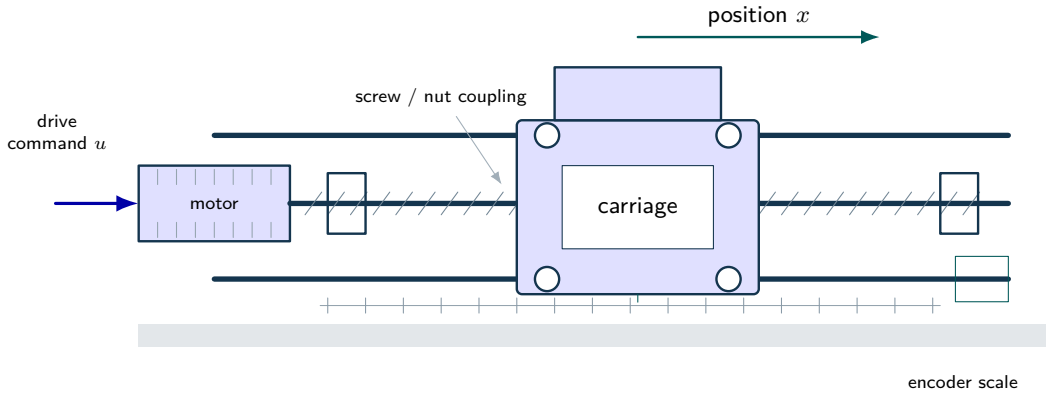
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

A normalized actuator has $G(s) = 1/[s(s+2)(s+6)]$ under unity negative feedback with proportional gain $K > 0$. Derive the strictly stable gain interval using Routh–Hurwitz, the imaginary-axis crossing and real-axis breakaway. Simulate $K = 12$, 60 and 100 to compare regulation and instability. A short time plot alone is not a stability test.

2 Physical system



Illustrative positioning axis; $G(s) = 1/[s(s+2)(s+6)]$ is stipulated

Command and readout
Drive command u and measured position x form a normalized teaching interface.

Stipulated dynamics
$G(s) = \dfrac{1}{s(s+2)(s+6)}$.
One integrator and two stable lags determine the pole geometry.

Geometry is illustrative
Screw pitch, motor constants, backlash and bearing friction are not inferred from the picture. Use the stated transfer function.

A motorized positioning axis illustrates the role of the integrator and two stable lags. The normalized transfer function is specified as a teaching model; the pictured dimensions do not identify its coefficients.

3 Characteristic equation

The unity-feedback denominator is

$$p(s) = s(s+2)(s+6) + K = s^3 + 8s^2 + 12s + K.$$

The first column of the Routh array is $1, 8, (96-K)/8, K$. Strict stability therefore requires $0 < K < 96$. At $K = 0$ there is a pole at the origin; $K = 96$ is a boundary, not a stable maximum.

4 Imaginary-axis crossing

At $K = 96$, $p(s) = (s+8)(s^2+12)$, so the crossing pair is $s = \pm j\sqrt{12}$. The remaining pole stays at -8.

5 Root-locus geometry

There are three open-loop poles and no finite zeros. The centroid is $-8/3$ and the asymptote angles are 60, 180 and 300 degrees. Real-axis locus segments are $(-\infty, -6)$ and $(-2, 0)$. Since $K(s) = -s(s+2)(s+6)$, a stationary point satisfies $3s^2 + 16s + 12 = 0$. The root $(-8 + 2\sqrt{7})/3 \simeq -0.9028$ belongs to $(-2, 0)$ and is the valid positive-gain breakaway. The other stationary point does not lie on a positive-gain real-axis segment.

6 Checking the plots

Use pole real parts to classify each gain before plotting. A finite horizon can conceal instability when an unstable real part is small. For stable positive gains, the integrator in the open loop gives zero constant-reference error, but says nothing about overshoot.

7 What to learn from the experiment

An algebraic stability interval provides a guardrail for numerical tuning. The endpoint K=96 must be excluded from strict asymptotic stability.

8 System sketch

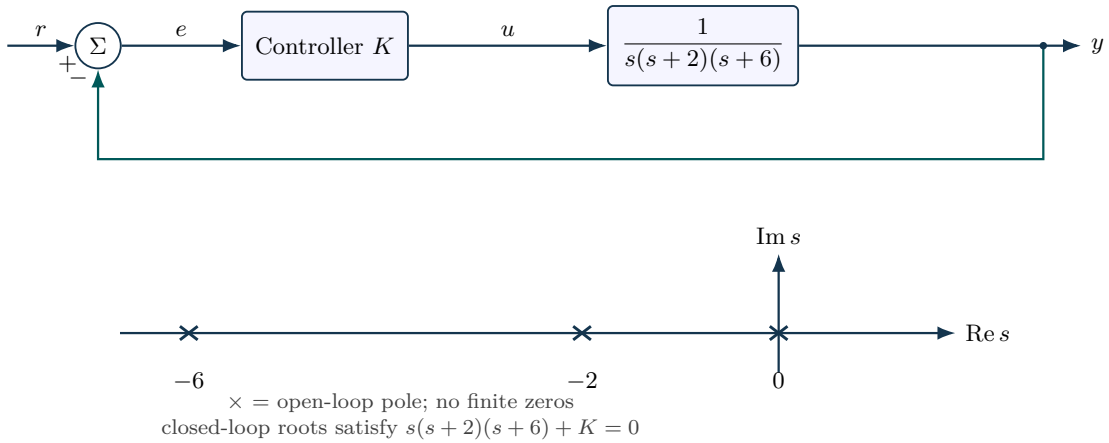
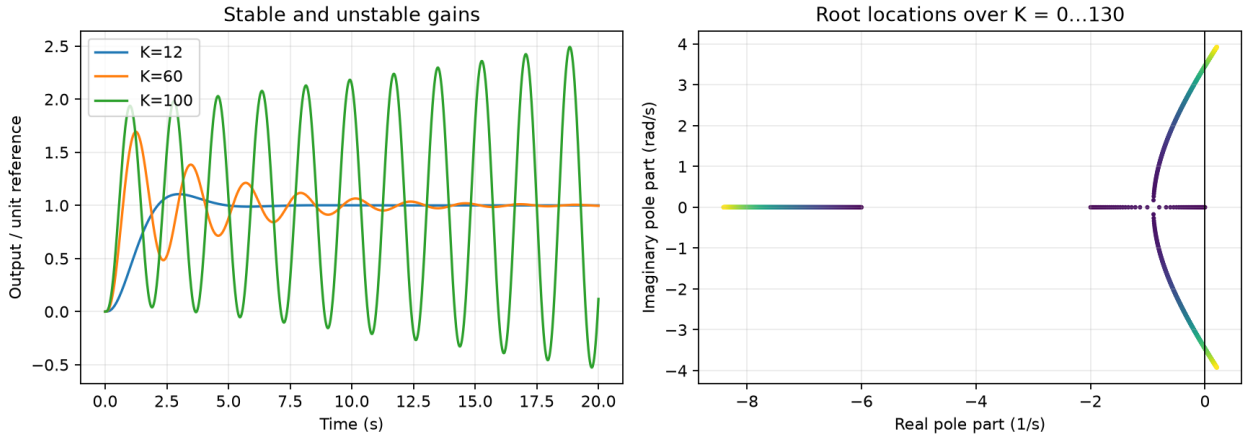


Figure 1. Unity negative feedback and the open-loop pole map. Pole symbols provide the starting geometry for the root-locus exercises.

9 Executed computational evidence

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K=12: largest pole real part=-0.788757
K=60: largest pole real part=-0.265547
K=100: largest pole real part=0.026030
Breakaway=-0.90283246; gain=5.04904248; boundary frequency=3.46410162 rad/s
```



Method references. The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.